

HOW MUCH FUTURE CONTEXT DOES STREAMING ACCENT CONVERSION NEED? A CONTROLLED STUDY USING CAUSAL PHONE TRANSLATION

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ABSTRACT

Streaming foreign accent conversion (FAC) systems choose a lookahead budget and report a single end-to-end latency, but no published work sweeps that budget or separates the delay a modelling choice imposes from the delay the hardware imposes. We train a causal phone translator at 16 lookaheads from 0 to 640 ms with every other factor held fixed, and report four results. First, quality improves smoothly with lookahead at -0.045 phone error rate (PER) per doubling ($R^2 = 0.98$) with *no locatable knee*: a piecewise fit is preferred under every axis treatment we tried, yet the breakpoint moves between 40 and 180ms and even the narrowest bootstrap interval spans 2.4 octaves. Instead of a knee we report a *saturation point* at ≈ 240 ms, beyond which each additional 20 ms frame buys less than the measured 2σ noise floor. A 40 ms budget, the value used by the current state of the art, forgoes $\approx 40\%$ of the achievable PER reduction. Second, a pre-registered hypothesis that vowels and approximants degrade faster than obstruents is *not supported*: all seven manner classes improve by 52–72% relative, and the pre-registered grouping explains less variance than the spread within its own groups. It still fails when errors are reweighted by measured acoustic cost, so this is not an artefact of PER charging every error equally. Third, canonical-phone conversion benefits more from future context than accent-faithful transcription at matched capacity (-0.045 against -0.029 PER per doubling, $1.55\times$), and within the canonical-phone arm the benefit is $1.32\times$ larger at positions where the speaker actually deviated from canonical. Fourth, per-chunk compute is flat in lookahead (-0.3% to $+1.9\%$ across a $17\times$ change in algorithmic latency, replicated on two architectures), which makes the two latency terms independently actionable. We release the harness and the per-condition outputs.

Index Terms— accent conversion, streaming speech, latency, lookahead, self-supervised models

1. INTRODUCTION

Streaming FAC exists and works. PHONOS [2] runs at ≤ 40 ms lookahead and reports an 81% reduction in non-native accent confidence. What does not exist is any account of *why* 40 ms, what it costs in quality, or what any of it does on hardware that is not a datacentre GPU.

Three gaps follow, and this paper addresses the first two directly.

Harness, causality proof, sweep configurations, analysis scripts with self-tests and per-condition hypothesis dumps: <https://github.com/DipeshTripathi13/streaming-fac-lookahead>. Data is L2-ARCTIC [1] via KoelLabs/L2Arctic (CC-BY-NC-4.0); that licence is non-commercial, so configurations are released rather than weights.

The budget is chosen, not characterised. PHONOS *states* 40 ms and publishes no ablation showing 40 ms is necessary or sufficient; TVTSyn fixes 4 future tokens (≈ 80 ms) without varying them. DarkStream does ablate, over three points $\{0, 140, 280\}$ ms, and finds the third buys < 1 pp, but three points scored on anonymisation accuracy cannot locate where returns stop, which is the question a deployment budget actually asks. “What would 160 ms buy you” still has no published answer.

Algorithmic and computational latency are reported fused. A figure such as “ < 241 ms on a single GPU” combines a modelling choice with unstated inference and buffering costs. A reader cannot tell whether a faster chip would help. We decompose throughout as

$$t_{e2e} = \underbrace{c + L}_{t_{\text{alg}}} + t_{\text{cmp}} + t_{\text{buf}}, \quad (1)$$

where c is chunk size, L is lookahead, t_{cmp} is per-chunk compute and t_{buf} is I/O and jitter buffering. t_{alg} is inherent: it is the delay on an infinitely fast machine, because the model refuses to emit frame n until it has seen frame $n+L$. t_{cmp} is contingent, quantise, prune, buy a faster chip.

Degradation is reported in aggregate. Existing results are utterance-level MOS, WER or accentedness. Which *sounds* break first is unreported.

Our contributions:

1. A latency–quality characterisation of streaming FAC across 16 lookahead budgets, reporting an exchange rate and a floor-relative saturation point rather than an unidentifiable knee (§5.2).
2. A negative result on a pre-registered phoneme-class hypothesis (§5.3).
3. Evidence that accent conversion needs more lookahead than accent-faithful transcription at matched capacity (§5.4).
4. Two patches that make a masked SSL encoder actually causal, with a truncation proof (§3.2).

2. RELATED WORK

Streaming FAC is established. Nguyen et al. [3] present a streaming non-autoregressive accent-conversion system on an Emformer encoder, demonstrating feasibility at stable latency; PHONOS [2] subsequently targets much lower latency with bounded future context. Neither characterises how quality varies with that context, which is the axis we isolate. PHONOS, TVTSyn [4] and DarkStream [5] come from one group and pair conversion with speaker anonymisation, and two report more than is sometimes credited. TVTSyn gives CPU as well as GPU latency (≈ 132 and ≈ 79 ms at 60 ms chunk), so a CPU number is not unpublished, a CPU latency–*quality* curve

Table 1. Truncation test on wavlm-base-plus. Relative L_2 change in earlier frames after deleting later audio.

configuration	relative L_2	causal?
attention mask only	1.14×10^{-2}	no
+ causal positional conv	6.00×10^{-3}	no
+ cumulative GroupNorm	2.60×10^{-2}	no
both patches	6.14×10^{-6}	yes

is. DarkStream ablates look-ahead over $\{0, 140, 280\}$ ms and selects 140 ms because the extra 140 ms buys <1 pp at double the buffering delay: a three-point ablation, scored on anonymisation accuracy rather than conversion quality, whose marginal return collapses in the same region where we place saturation on phone error rate over 16 budgets. That agreement across objectives is a stronger position than absent prior evidence would have been.

StreamVC [6] reports mobile operation without device, core or thread details, limiting hardware-level reproducibility. A survey [7] frames the space; streaming anonymisation [8, 9, 10] shares the constraint but not the accent objective; offline FAC [11, 12] and low-latency VC [13] bracket the design space. Our contribution is orthogonal: we characterise an axis these systems parameterise but do not study, and claim no quality improvement over any of them.

3. METHOD

3.1. A causal phone translator as the probe

Training a full FAC pipeline per lookahead is prohibitive. We instead isolate the component the research question is about. L2-ARCTIC [1] provides, per utterance, both the canonical phone sequence a native speaker would produce (*g2p*) and the sequence the non-native speaker actually produced (*ipa*). Accent conversion is then directly trainable as

$$\text{non-native audio} \rightarrow \text{native phone sequence},$$

a CTC head over a frozen causal encoder. This is the supervised core of the task, it costs a GPU-day rather than 200 GPU-hours, and swapping the target tensor from *g2p* to *ipa* yields a matched-capacity control that differs in *one tensor*, a tighter control than AC-versus-VC.

3.2. Making the encoder causal

Masking self-attention in wav2vec2/WavLM-base [14, 15] does *not* produce a causal encoder. Two leaks remain: the positional convolution (*pos_conv_embed*, depthwise, kernel 128, symmetric padding) admits 1.28 s of future at a 20 ms frame rate; and the feature-encoder GroupNorm normalises each channel over the entire utterance, making every output frame depend on every input frame.

We verify causality by truncation: delete audio after frame t and check whether any earlier frame moved (Table 1). Note the third row: patching GroupNorm alone makes matters *worse*. A partial causality fix is not a partial improvement.

3.3. Lookahead is quantised

Lookahead is realised as $\lceil L/f \rceil$ frames at frame rate $f = 20$ ms. A knee narrower than 20 ms is not merely unmeasured but *unrepresentable*, and sampling finer than f silently creates duplicate condi-

tions that reweight any fit. Our dense grid is therefore 20 ms, and we assert zero collisions in t_{alg} across all conditions before fitting.

4. EXPERIMENTAL SETUP

Frozen causal WavLM-base-plus, CTC phone head, 1200 steps, batch 8, chunk 40 ms, 100-frame lookback, speaker-disjoint test split of 400 utterances. Two sweeps: a 7-lookahead $\times$ 2-target $\times$ 3-seed sweep (42 runs) and a 16-lookahead $\times$ 2-seed dense sweep, run on *both* target arms (64 runs), all on an NVIDIA T4. Compute measurements use an Apple M4 and an ARM64 Linux host, CUDA-event or *perf_counter_ns* timed, 28–50 repetitions with three warm-up iterations discarded.

Noise floor. Two sweep processes briefly ran concurrently by operator error, repeating six conditions at *identical seed and configuration*. Those repeats are the only direct measurement of fixed-seed run-to-run noise available, and a multi-seed design cannot produce it: pairing within seed cancels seed effects but not this residual. Differences ranged 1.5×10^{-4} to 6.5×10^{-3} PER, giving $\sigma = 0.0022$ and a 2σ resolvable threshold of **0.0044** PER. Every effect below is gated on it.

5. RESULTS

5.1. Compute is flat in lookahead

Across $L \in [0, 640]$ ms at chunk 40 ms on the M4, t_{alg} spans 40–680 ms ($17\times$) while median per-chunk compute moves from 43.23 ms to at most 44.06 ms, a range of -0.3% to $+1.9\%$. On ARM64 the span is 0 to $+7.6\%$. The two terms of Eq. 1 are therefore independently actionable, which is the practical justification for reporting them separately.

Independently actionable is not the same as free. At this configuration $t_{\text{cmp}} \approx 43\text{--}44$ ms against a 40 ms chunk, i.e. $\text{RTF} = t_{\text{cmp}}/c \approx 1.08 > 1$: the base encoder does not sustain real time at 40 ms chunks on this machine, whatever L is. Eq. 1 describes the delay of a single chunk and must be read alongside the steady-state throughput constraint

$$\text{RTF} = t_{\text{cmp}}/c < 1, \quad (2)$$

which lookahead leaves untouched but chunk size and hardware do not. Changing L is close to free in compute; being able to run at all is not.

5.2. RQ1: an exchange rate and a saturation point, not a knee

Canonical-phone-arm PER falls monotonically from 0.4256 at $L=0$ to 0.1561 at $L=640$ ms. Fitting $L \geq 20$ ms on $\log_2 L$ gives

$$\Delta\text{PER} = -0.0452 \text{ per doubling}, \quad R^2 = 0.983, \quad (3)$$

over 15 points.

There is curvature, and it is not localisable. BIC prefers a two-segment fit under all three axis treatments we tried ($\Delta\text{BIC} = -22.2, -31.2, -36.8$), which by the usual thresholds [16] is strong evidence. But the breakpoint lands at 40, 180 and 180 ms respectively, and every bootstrap 90% interval spans 2.4 to 3.0 octaves. Sixteen points did not rescue a knee. The reason is mechanical and worth stating: on a $\log_2(L+1)$ axis the $L=0 \rightarrow 20$ ms gap is 4.39 units while every other gap is ≈ 1.0 , so a piecewise fit places a breakpoint just past that gap whatever the data does.

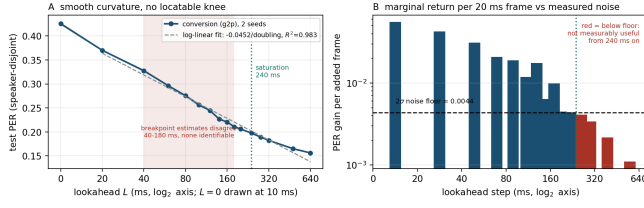


Fig. 1. Dense sweep, canonical-phone arm, 16 lookaheads $\times$ 2 seeds. **Left:** smooth curvature; the shaded band spans the breakpoint estimates produced by three defensible treatments of $L=0$, none identifiable. **Right:** marginal return per added 20 ms frame against the measured 2 σ floor; red bars are below it.

Table 2. Canonical-phone-arm PER against lookahead (mean of 2 seeds) and marginal return per added 20 ms frame, in units of the measured 2 $\sigma=0.0044$ floor. Below 1.0 $\times$ an additional frame is not measurably useful.

L (ms)	PER	gain/frame	$\times$ floor
0	0.4256	—	—
20	0.3701	0.0556	12.7
40	0.3275	0.0426	9.8
60	0.2963	0.0312	7.2
80	0.2754	0.0209	4.8
100	0.2564	0.0190	4.4
120	0.2446	0.0118	2.7
140	0.2269	0.0176	4.0
160	0.2206	0.0064	1.5
180	0.2106	0.0099	2.3
200	0.2062	0.0045	1.0
240	0.1973	0.0044	1.0
280	0.1891	0.0041	0.9
320	0.1823	0.0034	0.8
480	0.1649	0.0022	0.5
640	0.1561	0.0011	0.3

What we report instead is the region beyond which marginal return falls below the measured floor. The grid is 20 ms up to 200 ms and coarser above it, so past that point we report the *average* PER reduction per 20 ms equivalent of added lookahead over each sampled interval, not per individually observed 20 ms step. On that basis gains run +0.056 (0 $\rightarrow$ 20 ms), +0.021 (60 $\rightarrow$ 80), +0.0045 (180 $\rightarrow$ 200), and fall below the 0.0044 floor for every sampled interval beyond 240 ms. We therefore report a **saturation region of 200–280 ms** rather than a point: the 140 $\rightarrow$ 160 and 180 $\rightarrow$ 200 ms steps are already only 1.0–1.5 $\times$ the floor, the floor itself is estimated from six repeats, and the intervals above 320 ms average over 160 ms of lookahead. Unlike a knee, this quantity depends on the floor rather than on an axis choice.

The cost of a 40 ms budget. PER at $L=40$ ms is 0.3275; at 240 ms it is 0.1973. **Relative to the 40 ms condition, increasing lookahead to 240 ms reduces PER by a further 39.8%**, and to 640 ms by 52.3%. Measured instead against the full range available here ($L=0 \rightarrow 240$ ms), a 40 ms budget has already captured 57% of the attainable reduction and leaves 43% unclaimed. “Current budgets are under-provisioned” is thus supportable as diminishing-returns geometry with a measurable saturation point, not as a knee.

Table 3. Relative error-rate reduction, $L=0 \rightarrow 640$ ms. B = pre-registered as breaking first, S = as surviving. The predicted grouping does not order the data.

	class	rel. gain	ref. phones
S	nasal	0.716	4 020
B	approximant	0.700	4 575
S	fricative	0.687	7 659
B	vowel (mono)	0.673	13 008
B	vowel (diph)	0.654	4 824
S	stop	0.615	7 071
S	affricate	0.525	1 098

5.3. RQ2: manner class is the wrong axis

We pre-registered, before any of this data existed, that vowels and approximants would break first (formant trajectories over 80–250 ms) while stops, fricatives, affricates and nasals would survive short lookahead (locally determined gestures). Testing this on 42,255 unique reference-phone tokens evaluated across seven lookahead conditions (295,785 phone-condition observations), with each substitution or deletion charged to the class of the *reference* phone:

Group means differ in the predicted direction (0.676 vs 0.636), but that is the weakest available reading and three stronger checks fail. The effect size is 0.66: the between-group difference (0.040) is smaller than the pooled SD across classes (0.060). The surviving group’s *internal* spread is 0.190, 4.7 $\times$ the between-group difference. And 5 of 12 pairwise orderings are violated, with nasals out-gaining all three breaks-first classes. **H2 is not supported.** The finding is not that lookahead fails to help vowels, every class improves 52–72% relative, all above the floor. The finding is that manner class does not predict which sounds need right context.

Reweighting each error by its acoustic cost does not rescue it. PER charges every error 1.0, so the natural objection is that a substituted vowel and a substituted stop cannot sound equally wrong. We force-align each canonical phone string to its own synthesis with an espeak-IPA CTC model, DTW-align the hypothesis synthesis to it, and charge the spectral distortion inside each phone’s span to that phone’s class. The measure is validated first: wrong phones carry 1.72 $\times$ the distortion of correct ones (86.8 vs 50.5).

The verdict is unchanged. Acoustic weighting does move in H2’s favour, effect size 0.607 $\rightarrow$ 0.756 and ordering violations 4/12 $\rightarrow$ 2/12 against the sequence metric on the same utterances, but not far enough to clear the pre-registered threshold of 1.0, and the failure has the same shape as before: the “survives” group’s internal spread (0.120) is 3.5 $\times$ the difference between the groups (0.034), and a nasal again out-gains a diphthong. So “PER mis-weights the classes” is no longer an available explanation for the sequence-level refutation.

One explanation does remain open, and it is not the one this test closes. Both signals here are TTS renderings of phone strings, so the synthesiser produces canonical formant trajectories for whatever string it is given. This measures the cost of getting a phone’s *identity* wrong; it says nothing about a wrong trajectory for a phone whose identity is right. That still needs real converted audio against a native rendition (§??). The sample is also far smaller than the sequence-level test, 1451 phones per condition against 295,785, and is weighted accordingly.

Table 4. Both arms on the dense grid, mean of 2 seeds. The gap is the transcription arm’s PER minus the canonical-phone arm’s: near zero at $L=0$ and widening monotonically with lookahead.

L (ms)	conversion	transcription	gap
0	0.4256	0.4427	+0.0171
20	0.3700	0.3896	+0.0196
40	0.3275	0.3600	+0.0325
80	0.2754	0.3271	+0.0517
160	0.2206	0.2908	+0.0702
240	0.1973	0.2755	+0.0782
320	0.1823	0.2666	+0.0843
640	0.1561	0.2501	+0.0939
exchange rate (PER/doubling)			
conversion	−0.0452	$R^2 = 0.983$	
transcription	−0.0292	$R^2 = 0.984$	

5.4. RQ3: conversion needs more lookahead than transcription

Identical architecture, capacity, data order, seed and step count; the two arms differ only in the target tensor. Endpoint gain $L=0 \rightarrow 640$ ms is -0.2695 (63.2% relative) for conversion against -0.1893 (42.8%) for accent-faithful transcription, a paired difference of $+0.0802$ (sd = 0.0035, $t = 39.7$).

The dense grid replicates this, and sharpens it. Running the full 16-lookahead $\times$ 2-seed sweep on the transcription arm as well puts both arms on the same dense grid (Table 4). The endpoint gains reproduce the 7-point result almost exactly, 63.3% for conversion against 43.5% for transcription, against 63.2% and 42.8% before, on an independent, denser sample. The exchange rates separate cleanly:

conversion -0.0452 vs transcription -0.0292 PER per doubling,

$R^2 = 0.983$ and 0.984 respectively, so **conversion buys $1.55\times$ as much per doubling of lookahead.**

The mechanism is visible in the gap rather than the levels. At $L=0$ the two arms are nearly the same task (0.4256 vs 0.4427, a gap of 0.0171); by $L=640$ ms the gap is 0.0939, **$5.5\times$ wider**. It widens in 13 of 15 adjacent steps, and the transcription arm is worse at 16 of 16 lookaheads, every one by more than the 0.0044 floor.

The arms differ in their target tensor, which carries more than the task: label entropy, sequence length and annotation noise move with it. We therefore repeat the test *inside* the canonical-phone arm, where none of those can differ. Splitting reference positions by whether the speaker actually deviated from canonical (g2p $\neq$ ipa), the exchange rate is -0.0520 PER per doubling at accent-changing positions against -0.0394 at unchanged ones, a factor of **1.32** from the same model on the same utterances. That converges with the $1.55\times$ between arms. One caveat belongs with it: in *relative* terms the ordering reverses (51% against 71%), because accent-changing positions start far harder (0.590 against 0.354) and the gap between the two widens with lookahead. Future context buys more absolute accuracy where accent lives, but does not close that gap.

Two secondary points fall out. Both arms saturate at the same ≈ 240 ms, and neither yields a locatable knee (breakpoints move to [40, 240, 160] ms across axis treatments for transcription, against [40, 180, 180] for conversion), so §5.2’s structural conclusion is not an artefact of the canonical-phone arm. And the transcription arm’s own curve is just as log-linear, so the shallower slope is a smaller effect of lookahead, not a noisier measurement of the same one.

The claim is carried by the *preference margin*, PER against the other arm’s labels minus PER against its own, which grows monotonically in 3/3 seeds for conversion ($+0.032$ at $L=0$ to $+0.099$ at 640 ms; 18/18 adjacent step $\times$ seed comparisons positive) while the transcription control is 0/3 monotone and drifts *down* (-0.0103). Building the claim on a difference rather than a level turned out to matter (§6).

5.5. t_{buf} : jitter measured, I/O not

Two 30 s duplex runs on the M4 at 40 ms blocks give a required jitter buffer of 0.14 and 0.08 ms with zero over/underruns, so the jitter term is negligible. The I/O term is **not measured**: an acoustic loopback is defeated by platform echo cancellation, which scales with its own reference and so cannot be overcome by a louder probe, and the driver-reported figure is queueing rather than converter delay (its excess over the requested block is *identically* 201.69 ms at both 10 and 20 ms blocks). We report the gap rather than substituting a number.

6. A MEASUREMENT FAILURE WORTH REPORTING

A padding bug rewrote the curve’s shape. A masked block handed a zero-padded hidden-state tensor to a depthwise convolution and feed-forward that do not respect the key-padding mask. The correction was not a constant offset. It grows with lookahead, from -0.014 PER at $L=0$ to -0.086 at 320 ms, $16.4\times$ the noise floor, so the buggy curve *understated* the benefit of lookahead, biasing the central quantity of RQ1. A constant offset would have cancelled in every within-run comparison; this did not. Every number reported here was regenerated after the fix. Two further failures, a breakpoint estimator that returns a breakpoint whether or not one exists and an alphabet mismatch that silently bucketed 57% of reference tokens, are documented in the repository.

7. LIMITATIONS

Every quality result here is phone-level, not perceptual, and this is a phone-translation probe rather than an end-to-end FAC system: the latency–accuracy relation reported here need not transfer to synthesis, prosody, speaker similarity or perceived accentedness. “Unfunded” would be the wrong reason to give for the missing listening test, stimuli built from these outputs cannot carry the judgement, because the phone sequences hold no word or stress structure and a system at 0.16–0.42 PER emits no recognisable words; an informal session confirmed both. Training is 1200 steps at fixed budget, not to convergence, over a frozen encoder; the dense sweep has two seeds per arm and no repeated cells, so its noise floor is imported from six accidental repeats in the other sweep and carries no confidence interval of its own. t_{buf} ’s I/O term is unmeasured, and compute is characterised on Apple Silicon and ARM64 Linux only.

8. CONCLUSION

For causal phone translation, future context buys accuracy at an approximately log-linear average rate of ≈ 0.045 PER per doubling, with no identifiable knee and measurable returns to a saturation region of 200–280 ms, well beyond PHONOS’s 40 ms translator budget, relative to which a further 39.8% reduction is available. Lookahead changes algorithmic latency rather than per-chunk cost, though throughput must still satisfy $t_{\text{cmp}} < c$ (Eq. 2) to run at all.

9. ACKNOWLEDGEMENTS

This work received no external funding and used no institutional compute: all GPU results are from free-tier Google Colab (Tesla T4) and all CPU results from consumer hardware. Thanks to the maintainers of L2-ARCTIC, WavLM, sherpa-onnx and Piper, whose public releases made a single-author study of this scope possible.

10. COMPLIANCE WITH ETHICAL STANDARDS

This work used only the publicly released L2-ARCTIC corpus and involved no new data collection and no human subjects, so no ethical approval was required. No listening study was run, and no claim about perceived accentedness or naturalness is made. The author declares no competing interests.

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